Journal of Neuro-Oncology **66:** 361–363, 2004.

Clinical Study

Unusual metastatic features in a patient with concomitant malignant orbital melanoma and prostate carcinoma

Catherine Ars[1], Lionel Duck[1], Jean-Francois Baurain[1], Pierre Moulin[2], Thierry Duprez[3] and Jean-Pascal Machiels[1]
[1]*Medical Oncology Unit,* [2]*Department of Pathology,* [3]*Department of Radiology, Centre du Cancer, Université Catholique de Louvain, Brussels, Belgium*

Key words: ocular melanoma, pituitary metastases, prostate carcinoma, SIADH

Summary

A 73-year-old man with a history of malignant orbital melanoma and prostate carcinoma was admitted for progressive visual disturbance. Brain magnetic resonance imaging showed a suprasellar enhancing nodular lesion with major impingement on the anterior optical ways and sellar invasion. The extensive imaging work-up could not demonstrate with certainty its origin. Surprisingly, the transphenoidal biopsy of this patient revealed a prostate cancer metastasis outlining the importance of a histopathological diagnosis of cerebral metastases in patients with multiple malignancies when there is a doubt about the nature of the lesion.

Case report

A 73-year-old man with a history of double malignancy was admitted for progressive visual disturbance. He had been treated six years previously for a left intraocular melanoma. Conservative treatment with ophthalmic radioactive applicator using iodine-125 was initially performed but local relapse after two years required enucleation. At the time of orbital relapse, prostatic adenocarcinoma with para-aortic node metastases was diagnosed. Androgen deprivation allowed disease control for three years, until the appearance of supra-clavicular adenomegalies. He then received unsuccessfully estramustine and thereafter paclitaxel (weekly administration), which allowed disease stabilization till admission.

Right temporal hemianopsia confirmed at visual field examination was the only complaint at admission. Brain magnetic resonance imaging showed a suprasellar enhancing nodular lesion of 30 mm in greatest diameter with major impingement on the anterior optical ways and sellar invasion (Figure 1A and B). CT scan of the skull base with bone window viewing failed to reveal any osseous involvement with the tumor (not illustrated). A contiguity metastasis from orbital origin rather than a prostatic one was thus far suspected. Whole-body CT scan work-up revealed multiple liver metastases together with supra-clavicular and cervical adenomegalies. No metastases were detected at bone scintigraphy. Blood analysis and endocrine evaluation pointed out only a mild hypothyroidism (free-T4: 0.7 ngr dl^{-1}, normal value: 0.8–2.0 ngr dl^{-1}) with no other relevant abnormalities. A trans-sphenoidal biopsy of the supra-clavicular mass demonstrated metastatic tissue from prostate adenocarcinoma (Figure 2). Ten days later, the patient developed a severe hyponatremia (sodium: 113 meq l^{-1}, normal value: 135–145 meq l^{-1}) in the absence of edema or dehydration. Urine osmolality was high (657 mOsm kg^{-1}) and plasma osmolality low (272 mOsm kg^{-1}, normal value: 280–300). Injection of 100 mg of hydrocortisone did not improve the natremia. A syndrome of inappropriate antidiuretic hormone secretion (SIADH) was suspected and the hyponatremia resolved with hydric restriction. Whole brain radiation (20 grays) was delivered with transient visual improvement. Two months later, the patient was re-admitted and died of systemic cancer progression.

Discussion

Prostate adenocarcinoma commonly metastasizes to lymph nodes and bones with occasional visceral

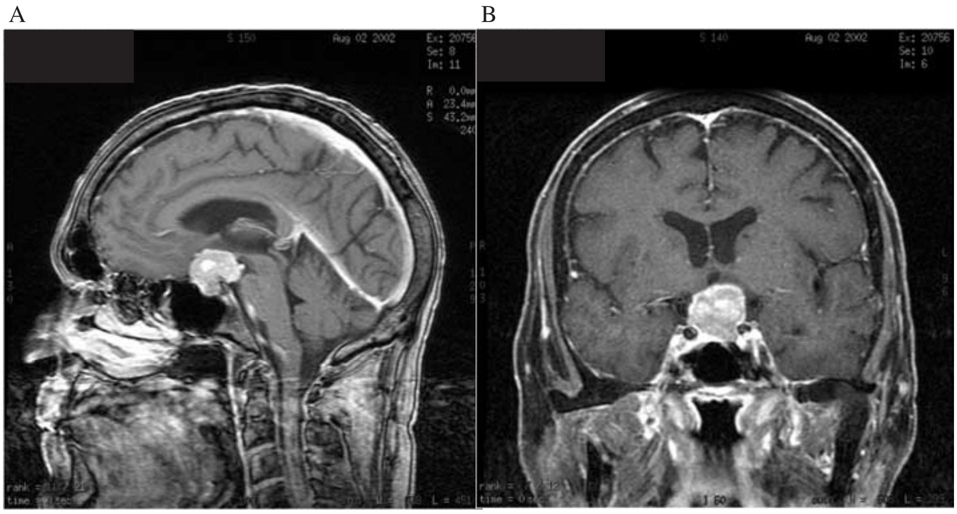

Figure 1. Contrast-enhanced T1-weighted spin-echo MR views in the sagittal (A) and coronal (B) planes showing a nodular suprasellar mass with major impingement on the optic chiasma and intra-sellar invasion.

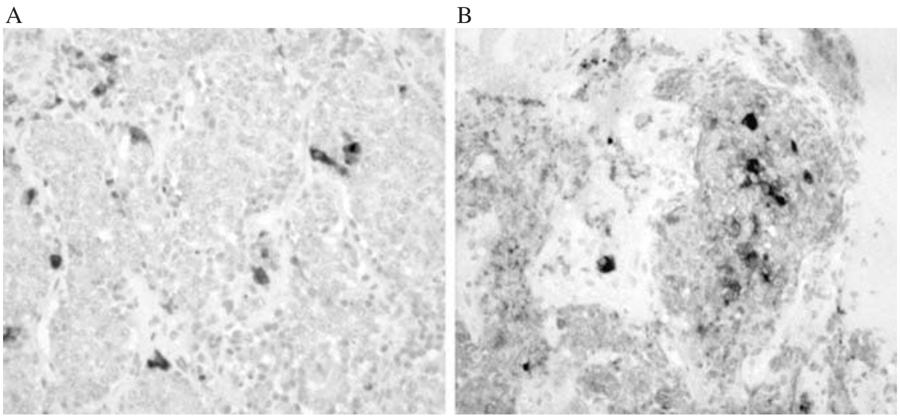

Figure 2. (A) Islets of neoplastic epithelial cells with interspaced endocrine cells labeled with anti-synaptophysin antibody. (B) Neoplatic cells display a medium intensity labeling for prostatic acid phosphatase

deposits. Brain metastases occur in less than 1% of patients in autopsy series and the majority of these metastases are asymptomatic. The symptomatic ones in the sellar region are exceptional with only a few cases reported [1,2]. The prognosis of these patients depends mainly on the stage of the disease. A favourable response to combined androgen blockade has been reported for one patient with no prior hormonotherapy [2]. However, for hormone refractory patient, the prognosis is dismal due to poor efficacy of available therapy. In addition, our patient developed severe hyponatremia compatible with a syndrome of inappropriate secretion of ADH (SIADH). Only nine cases of SIADH associated with prostate adenocarcinoma have been previously reported [3]. However, we were unable to demonstrate an elevated level of ADH in the cancer tissue of the transphenoidal biopsy (data not shown). Although speculative, another explanation for the hyponatremia is that the tumor caused irritative stimulation of ADH release from neurohypophysis as sometimes observed with intracranial lesions such as trauma, infection or vascular accident.

In this particular case, we biopsied the suprasellar mass because the extensive imaging work-up could not demonstrate with certainty its origin. Surprisingly, the transphenoidal biopsy of this patient revealed a prostate cancer metastasis outlining the importance of a pathological diagnosis of intra-cranial metastases in patients with multiple malignancies when there is a doubt about the nature of the lesion.

References

1. Couldwell WT, Chandrasoma PT, Weiss MH: Pituitary gland metastasis from adenocarcinoma of the prostate. Case Report. J Neurosurg 71: 138–140, 1989
2. Losa M, Grasso M, Giugni E, Mortini P, Acerno S, Giovanelli M: Metastatic prostatic adenocarcinoma presenting as a pituitary mass: shrinkage of the lesion and clinical improvement with medical treatment. Prostate 32: 241–245, 1997
3. Yamazaki T, Suzuki H, Tobe T, Sekita N, Kito H, Ichikawa T, Akakura K, Igarashi T, Ito H: Prostate adenocarcinoma producing syndrome of inappropriate secretion of antidiuretic hormone. Int J Urol 8: 513–516, 2001

Address for offprints: Jean-Pascal Machiels, Medical Oncology Unit, Université Catholique de Louvain, Cliniques Universitaires Saint-Luc 10, avenue Hippocrate 1200 Brussels, Belgium; Tel.: 32-2-764 54 85; Fax: 32-2-764 54 28; E-mail: jean-pascal.machiels@onco.ucl.ac.be